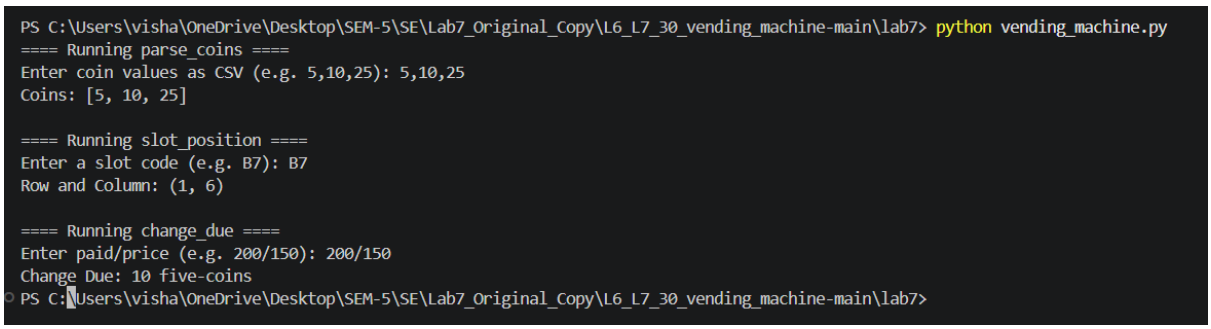
LAB 7 – FUZZING FOR INPUT VALIDATION BUGS

NAME : VISHAL

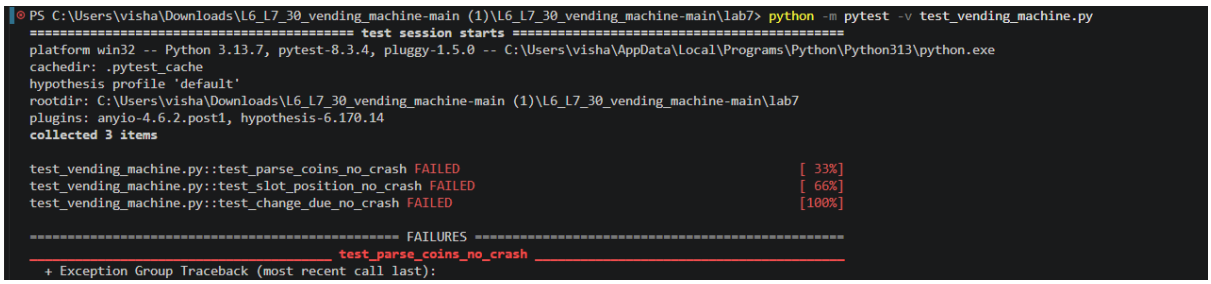
SRN: PES2UG24CS591

SECTION : J

1. Screenshot 1 – Original Program Execution



2. Screenshot 2 – Hypothesis Fuzzing Failures



3. Screenshot 3 – Fixed Python Code

File Edit View

```
def parse_coins(coins_csv):
    if not isinstance(coins_csv, str) or not coins_csv.strip():
        return []

    try:
        coins = [int(c.strip()) for c in coins_csv.split(',')]
        if any(c <= 0 for c in coins):
            return []
        return coins
    except ValueError:
        return []


def slot_position(code):
    if not isinstance(code, str):
        return None

    code = code.strip().upper()

    if len(code) < 2 or not code[0].isalpha() or not code[0].isascii() or not code[0].isalpha():
        return None

    if not code[1:].isdigit():
        return None

    column = int(code[1:])

    if not ('A' <= code[0] <= 'Z') or column < 1:
        return None

    row = ord(code[0]) - ord('A')
    return row, column - 1


def change_due(payment):
    if not isinstance(payment, str):
        return "invalid payment"

    parts = payment.split('/')

    if len(parts) != 2:
        return "invalid payment"

    try:
        paid = int(parts[0].strip())
        price = int(parts[1].strip())
    except ValueError:
        return "invalid payment"

    change = paid - price

    if change < 0 or change % 5 != 0:
        return "invalid payment"

    return str(change // 5) + ' five-coins'
```

4. Screenshot 4 – All Tests Passed

```
PS C:\Users\visha\Downloads\L6_L7_30_vending_machine-main (1)\L6_L7_30_vending_machine-main\lab7> python -m pytest -v test_vending_machine.py
========================================= test session starts =========================================
platform win32 -- Python 3.13.7, pytest-8.3.4, pluggy-1.5.0 -- C:\Users\visha\AppData\Local\Programs\Python\Python313\python.exe
cachedir: .pytest_cache
hypothesis profile 'default'
rootdir: C:\Users\visha\Downloads\L6_L7_30_vending_machine-main (1)\L6_L7_30_vending_machine-main\lab7
plugins: anyio-4.6.2.post1, hypothesis-6.170.14
collected 4 items

test_vending_machine.py::test_parse_coins_no_crash PASSED                                    [ 25%]
test_vending_machine.py::test_slot_position_no_crash PASSED                                  [ 50%]
test_vending_machine.py::test_change_due_no_crash PASSED                                     [ 75%]
test_vending_machine.py::test_valid_inputs PASSED                                            [100%]

========================================== 4 passed in 0.70s ==========================================
PS C:\Users\visha\Downloads\L6_L7_30_vending_machine-main (1)\L6_L7_30_vending_machine-main\lab7>
```

5. Screenshot 5 – Fixed Program Execution

```
PS C:\Users\visha\Downloads\L6_L7_30_vending_machine-main (1)\L6_L7_30_vending_machine-main\lab7> python vending_machine.py
==== Running parse_coins ====
Enter coin values as CSV (e.g. 5,10,25):


                                    5,10,25
Coins: [5, 10, 25]

==== Running slot_position ====
Enter a slot code (e.g. B7): B7
Row and Column: (1, 6)

==== Running change_due ====
Enter paid/price (e.g. 200/150): 200/150
Change Due: 10 five-coins
PS C:\Users\visha\Downloads\L6_L7_30_vending_machine-main (1)\L6_L7_30_vending_machine-main\lab7>
```